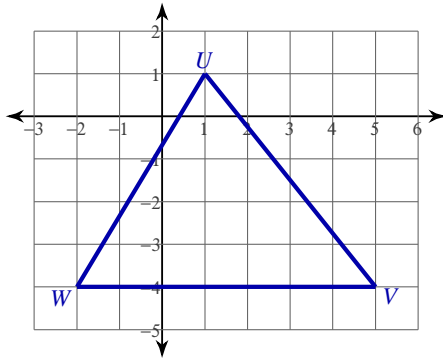
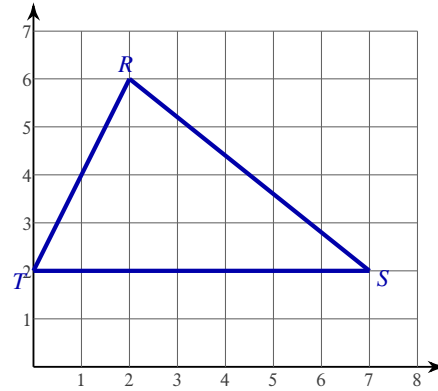


7)



8)



Find the coordinates of the centroid of each triangle given the three vertices.

9) $C(-7, -6)$, $B(-8, 0)$, $A(0, 0)$

10) $S(0, -6)$, $R(-10, -6)$, $Q(1, 0)$

11) $T(0, 0)$, $U(1, 3)$, $V(-2, 0)$

12) $Y(3, 4)$, $X(-5, 0)$, $W(0, 0)$

13) $E(6, -3)$, $D(-5, -3)$, $C(6, -1)$

14) $F(-7, -3)$, $G(-7, -5)$, $H(-3, -3)$

Critical thinking questions:

15) Two vertices of a triangle are $(0, 0)$ and $(9, 0)$. The centroid is $(6, 1)$. Find the third vertex of the triangle.

16) For question #1, connect the midpoints of each side of the triangle to form a smaller triangle within the original triangle. Find the coordinates of the centroid of the smaller triangle. What happened and why?